DYNAMIC EXPECTED-LOSS Optimization in Credit Risk

A Chronological, Explainable, and Proxy-Fair ML Framework

THESIS 2.0 AT A GLANCE

QUESTION

- How can credit-risk ML move beyond static classification into a calibrated Expected Loss decision framework?
- How should drift, financial consequences, explainability, and proxy-group fairness be evaluated together?

DATA

- Lending Club (Resolved 36m Loans) with 603,589 resolved loans.
- Target: final-status charge-off outcome.
- 21 baseline pre-origination features; repayment outcomes isolated for retrospective evaluation.

DESIGN

- Chronological split: Train 2011–2013, Validate & Recalibrate 2014, Frozen OOT Test 2015.
- Historical policy simulation rather than a live production deployment.

METHODS PIPELINE

1. Chronological Split
2. Fit Raw LightGBM
3. Isotonic Calibration
4. Historical LGD + EAD Proxy
5. Optimize EL Boundary
6. Audit Drift, SHAP & Fairness

RESULTS SUMMARY

MODEL PERFORMANCE

- LightGBM achieved OOT ROC-AUC = 0.6625, compared with 0.6569 for Logistic Regression.
- The performance gain was modest but consistent.
- Probability calibration required separate assessment beyond ranking performance.

EXPECTED LOSS POLICY

- A 5.00% projected principal-loss constraint selected the calibrated-PD boundary of 0.1696.
- Applied unchanged to 2015, the frozen policy retained 68.92% of originated loans.
- The retained portfolio realized 5.27% principal loss and 7.18% return on funded amount.

AUC COMPARISON

0.6569 Logistic Regression

0.6625 LightGBM

PORTFOLIO COMPARISON

6.35% Accept-All Return

7.18% EL Policy Return

0.1696 PD Approval Threshold

5.00% Projected EL Limit

5.27% Realized Loss Rate

1.8895 Renter / Non-renter FPR Ratio

DRIFT & EXPLAINABILITY

- Kitagawa decomposition indicated that the observed charge-off increase was dominated by within-segment deterioration rather than portfolio-mix change.
- TreeSHAP identified key contributors to the raw LightGBM output, including FICO, annual income, DTI, and related risk variables.

PROXY-GROUP FAIRNESS AUDIT

- Under the frozen policy, renters had a higher False Positive Rate than non-renters (38.5% vs 20.4%).
- The resulting FPR ratio was 1.8895.
- Renter status was used only as an illustrative socioeconomic proxy, not as a protected-class attribute.

TAKEAWAY

- Expected Loss optimization improved retrospective portfolio selection relative to an accept-all benchmark.
- Stable discrimination did not guarantee stable calibration over time.
- Credit-risk ML should be evaluated jointly on discrimination, calibration, temporal stability, financial consequences, explainability, and proxy-group fairness.

LIMITATIONS

- Originated loans only (selection bias).
- Historical LGD and funded amount used as simplified proxies.
- Home ownership is an illustrative socioeconomic proxy.
- Retrospective design with delayed-label assumptions.

Reproducibility: Python 3.13.5 | scikit-learn 1.6.1 | LightGBM | shap 0.50.0 | seed = 42

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